

Roll No.

TME-501

B. TECH. (ME) (FIFTH SEMESTER) MID SEMESTER EXAMINATION, Oct., 2023

HEAT TRANSFER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Discuss any combined phenomenon of conduction, convection, and radiation. (CO1)

OR

- (b) Derive 3D-conduction equation in cartesian coordinates and also discuss special cases. (CO1)
2. (a) Derive the relation for heat conduction through a cylinder and explain the concept of logarithmic mean area. (CO1)

OR

- (b) Derive the relation for heat conduction through a hollow sphere. (CO1)
3. (a) What is the critical thickness of insulation ? Derive the relation for cylinder and sphere. (CO1)

P. T. O.

(2)

OR

- (b) A hollow cylinder with an inner radius of 100 mm and an outer radius of 200 mm. Heat is applied at the inner surface at a constant rate of $1.16 \times 10^5 \text{ W/m}^2$ and the outer surface is maintained at 30°C . The thermal conductivity of cylinder material is 50 W/mK . Find the temperature of the inner surface. (CO1)
4. (a) Derive the expression for rectangular fin in case of heat dissipation from FIN insulated at the tip. (CO2)

OR

- (b) A straight fin of rectangular profile is constructed of stainless steel and has a width of 5 cm and thickness of 2.5 cm. The base temperature is maintained at 100°C and fin is exposed to a convection environment at 20°C with $h = 47 \text{ W/m}^2 \text{ K}$. Calculate the heat loss by the fin of one-meter length. (CO2)
5. (a) Explain in detail the efficiency and effectiveness of fin. (CO2)

OR

- (b) A motor body is 360 mm in diameter (outside) and 240 mm long, its surface temperature should not exceed 55°C when dissipating 340 W. Longitudinal fins 15 mm thick and 40 mm high are proposed. The convection coefficient is $40 \text{ W/m}^2\text{C}$. Determine the number of fins required. Atmospheric temperature is 30°C and the thermal conductivity is $40 \text{ W/m}^\circ\text{C}$. (CO2)

Roll No.

TME-502

B. TECH. (MECHANICAL ENGG.) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

DESIGN OF MACHINE ELEMENTS—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Explain the different steps followed in the design a machine element with a block diagram.
- (ii) What are the main influencing factors in the design of a machine component ? (CO1)

OR

- (b) List the different alloys added to steel. What is the effect of adding these alloys to properties of steel ? (CO1)
2. (a) Discuss the composition, classification and application of cast iron as engineering materials. (CO1, CO2)

OR

- (b) How are plain carbon steels, free cutting steels, alloy steels, high alloy steels are designated as per BIS specifications ? (CO1, CO2)

P. T. O.

(2)

3. (a) Define the following mechanical properties of Engineering materials : Strength, Elasticity, Plasticity, Stiffness, Resilience, Toughness, Ductility, Malleability, Brittleness, Hardness. (CO1, CO3)

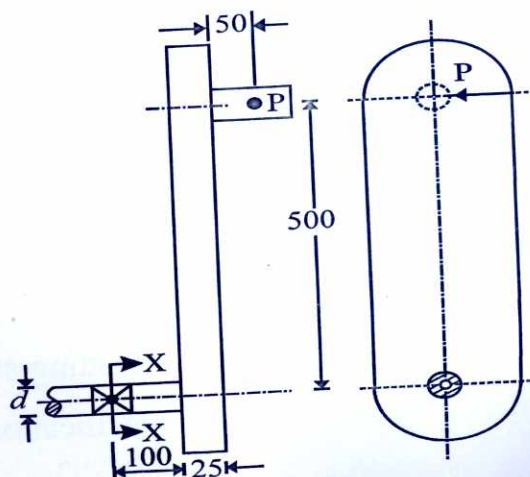
OR

- (b) It is required to design a cotter joint to connect two steel rods of equal diameter. Each rod is subjected to a tensile force of 50 kN. Design the joint and specify its main dimensions. The permissible tensile, crushing and shearing stresses are 70 MPa, 140 MPa and 35 MPa respectively. (CO1, CO3)

4. (a) It is required to design a knuckle joint to connect two circular rods subjected to an axial tensile force of 50 kN. The rods are co-axial and a small amount of angular movement between their axes is permissible. Design the joint and specify the dimensions of its components. The permissible tensile, crushing and shearing stresses are 80 MPa, 120 MPa and 40 MPa respectively. (CO3)

OR

- (b) The dimensions of an overhang crank are given in Fig. The force P acting at the crankpin is 1 kN. The crank is made of steel 30C8 (S_{yt} : yield tensile strength = 400 N/mm²) and the factor of safety is 2. Using maximum shear stress theory of failure, determine the diameter d at section XX. (CO3)



(3)

5. (a) A right angled bell-crank lever is to be designed to raise a load of 5 kN at the short arm end. The lengths of short and long arms are 100 and 450 mm respectively. The lever and the pins are made of steel 30C8 ($S_{yt} = 400 \text{ N/mm}^2$) and the factor of safety is 5. The permissible bearing pressure on the pin is 10 N/mm^2 . The lever has rectangular cross-section and the ratio of width to thickness is 3 : 1. The length to diameter ratio of fulcrum pin is 1.25 : 1. Calculate :

- (i) The diameter and the length of fulcrum pin;
- (ii) The shear stress in the pin;
- (iii) The dimensions of the boss of the lever at the fulcrum; and
- (iv) The dimensions of the cross-section of the lever.

Assume that the arm of bending moment on the lever extends up to the axis of the fulcrum. (CO3)

OR

- (b) Explain the different theories of failures in brief. (CO3)

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TME-503

B. TECH. (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
DYNAMICS OF MACHINES

Time : 1½ Hours

Maximum Marks : 50

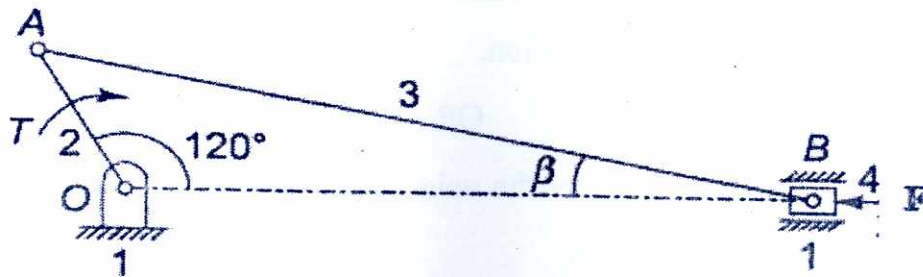
Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain the conditions for a body to be in equilibrium under the action of two forces, three forces and two forces and a torque. (CO1)

OR

- (b) What is mean by Piston effort and crank effort ? Explain in detail. (CO2)

2. (a) A slider-crank mechanism as shown in figure below is acted upon by a force $F = 200 \text{ N}$ at B. The dimensions are $OA = 50 \text{ mm}$, $AB = 200 \text{ mm}$. Determine the input torque T on the link OA to maintain the static equilibrium of the mechanism for the given configuration. (CO1)

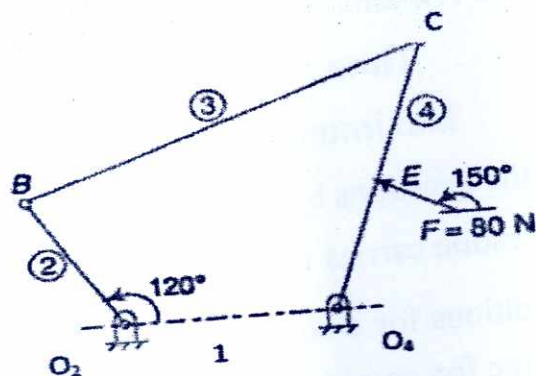


P. T. O.

(2)

OR

- (b) A four-link mechanism with the following dimensions is acted upon by a force of 80N as shown in figure below : $O_2O_4 = 500$ mm, $O_2B = 400$ mm, $BC = 1000$ mm, $O_4C = 750$ mm, $O_4E = 350$ mm. Determine the input torque T on the link O_2B for the static equilibrium of the mechanism for the given configuration. (CO1)



3. (a) What are turning-moment diagrams? Why are they drawn? (CO2)

OR

- (b) Drive the expression for the dimensions of flywheel rim. (CO2)
4. (a) The crank and connecting rod of a steam engine are 0.3 m and 1.5 m in length. The crank rotates at 180 rpm clockwise. Determine the velocity and acceleration of the piston when the crank is at 40 degrees from the inner dead centre position. Also determine the position of the crank for zero acceleration of the piston. (CO2)

OR

- (b) Drive the expressions for the velocity and acceleration of a piston in a slider crank mechanism. (CO2)

(3)

5. (a) The mass of flywheel of an engine is 6.5 tonnes and the radius of gyration is 1.8 m. It is found from the turning moment diagram that the fluctuation of energy is 56 kN-m. If the mean speed of the engine is 120 rpm, find the maximum and minimum speeds. (CO2)

OR

- (b) Define the terms 'coefficient of fluctuation of energy' and 'coefficient of fluctuation of speed', in the case of flywheels. (CO2)

Roll No.

TAE-501

B. TECH. (ME) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
ELECTRIC VEHICLE TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

(iii) Draw neat diagrams as necessary.

(iv) Make assumptions for any missing data.

1. (a) Give details of the timeline of Electric cars. (CO1)

OR

(b) Derive an expression for the tractive effort of an electric car. (CO1)

2. (a) What is meant by motor characteristic curve ? Explain with suitable example. (CO1)

OR

(b) "The use of an electric vehicle influences its components". Justify this statement. (CO1)

3. (a) What are the different types of cells ? Give the relative performance characteristics. (CO2)

P. T. O.

(2)

OR

(b) Explain the chemistry of the lead acid cell. (CO2)

4. (a) Explain the different battery parameters that need to be considered for use in electric vehicles. (CO2)

OR

(b) What is a Proton Exchange Membrane Fuel Cell ? Explain its construction and working. (CO2)

5. (a) Compare the different drive trains used in electric cars. (CO1)

OR

(b) Detail the benefits of Electric cars over IC engines. (CO1)

Roll No.

TRA-501

B. TECH. (ME) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
ROBOTICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define the terms robot and robotics. What are different laws of robotics for their ethical conduct ? (CO1)

OR

- (b) How to define the degrees of freedom of a robot manipulator ? Also, define the following types of manipulators : Spatial manipulator, Planar manipulator, and Redundant manipulator. (CO1)

2. (a) Discuss about the various robot arm configurations with neat sketches. (CO1)

OR

- (b) Discuss about the robot's end-effector with its different types. Also, define the world, joint and tool reference frames. (CO1)

P. T. O.

(2)

3. (a) How to evaluate the description of a position and the description an orientation ? (CO1 & CO2)

OR

- (b) Discuss about the mapping in general frames and describe the homogeneous transform. (CO1 & CO2)
4. (a) A frame {B} is rotated relative to another frame {A} about Z axis by 30 degrees and translated 6 units in X and 2 units in Y. The Z axis is pointing out of the page. Thus, we have a description of ${}^A_B T$. Find ${}^B_A T$. (CO2)

OR

- (b) A frame {Q} is rotated relatives to a frame {P} about Z axix by degrees; the Z axis is pointing out of the page. Position vector of a point with respect to frame {Q} is given by $[0 \ 2 \ 0]^T$. Calculate the position of that point with respect to frame {P}. (CO2)
5. (a) Define the following terms related to manipulators : (CO2 & CO3)
- (i) link-joint parameters
 - (ii) forward and inverse kinematics

OR

- (b) What do you understand by the D-H notation ? Write the different steps involved in D-H algorithm to derive the forward kinematic model. (CO2 & CO3)

Roll No.

DETF-02

B. TECH. (MECHANICAL ENGINEERING)

(FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

POWER PLANT ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) A power station has to supply load as follows : (CO4)

Time (hours)	Load (MW)
0-6	45
6-12	135
12-14	90
14-18	150
18-24	75

(i) Draw the load curve.

(ii) Draw load duration curve.

P. T. O.

- (iii) Choose suitable generating units to supply the load.
- (iv) Calculate the load factor.
- (v) Calculate the plant capacity factor.

OR

- (b) A 70 MW power station has an annual peak load of 45 MW. The power station supplies loads having maximum demands of 15 MW, 18 MW, 13 MW and 8 MW. The annual load factor is 0.42. (CO4)

Find :

- (i) Average load
- (ii) Energy supplied per year
- (iii) Diversity factor
- (iv) Demand factor

2. (a) Discuss the following terms : (CO1)

- (i) Peak load
- (ii) Load factor
- (iii) Diversity factor
- (iv) Demand factor

OR

- (b) Enumerate and explain briefly the straight line method used to calculate the depreciation cost. Define the following terms : (CO1)

- (i) Fixed cost
- (ii) Operational cost.

(3)

DETF-02

3. (a) Prove the following related to chimney height :

(CO3)

$$H_1 = H \left[\left(\frac{m_a}{m_a + 1} \right) \frac{T_g}{T_a} - 1 \right]$$

OR

- (b) Explain with the help of a neat diagram the arrangement of the Fluidised bed combustion (FBC) system.

(CO3)

4. (a) Calculate the mass of flue gases flowing through the chimney when the draught produced is equal to 1.9 cm of water. Temperature of flue gases is 290°C and ambient temperature is 20°C. The flue gases formed per kg of fuel burnt are 23 kg. Neglect the losses and take the diameter of the chimney as 1.8 m.

(CO2)

OR

- (b) Explain with neat sketches the construction and working of Benson boiler.

(CO2)

5. (a) Define the term steam nozzle. Discuss *two* functions of steam nozzle.

(CO5)

P. T. O.

(4)

DETF-02

OR

- (b) The maximum demand of a power station is 85000 kW and daily load curve is described as follows : (CO5)

Time (hours)	Load (MW)
0-6	35
6-8	55
8-12	65
12-14	52
14-18	75
18-22	86
22-24	40

- (i) Determine the load factor of power station.
- (ii) What is the load factor of stand by equipment rated at 25 MW that takes up all load in excess of 65 MW ? Also calculate its use factor.

Roll No.

DEEM-03

B. TECH. (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MANAGEMENT OF INTELLECTUAL PROPERTY RIGHTS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Provide an in-depth definition of a patent, exploring its legal, technical, and commercial dimensions. (CO3/CO4)

OR

- (b) Evaluate the multifaceted nature of Indian patent law and its implications on International patent protection. (CO3/CO4)

2. (a) Conduct an extensive analysis of what can be patented, considering various categories of inventions and their nuances. (CO3/CO4)

OR

- (b) Engage in a critical examination of the restrictions on patentability in India, citing specific sections of the Patent Act, 1970, and discussing their historical context and implications. (CO3/CO4)

P. T. O.

(2)

3. (a) Investigate and present a comprehensive overview of the diverse types of patent applications, including utility patents, design patents, and International applications. (CO4/CO5)

OR

- (b) Examine the strategic implications of choosing a specific type of patent application based on the nature of the invention and the intended markets. (CO4/CO5)
4. (a) Scrutinize the concept of early publication within the Patents Act, dissecting any relevant provisions and their legal, procedural, and strategic significance. (CO5/CO1)

OR

- (b) Discuss case studies and real-world scenarios where early publication played a pivotal role in patent strategy. (CO5/CO1)
5. (a) Delve deep into the qualifications, responsibilities, and ethical considerations for individuals aspiring to become patent agents. (CO1/CO2)

OR

- (b) Explore the complex web of rights, responsibilities, and privileges that patent agents possess in the patent application and prosecution process. (CO1/CO2)

Roll No.

DEDE-04

B. TECH. (ME) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

PYTHON FOR DATA SCIENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

(iii) Draw neat diagrams as necessary.

(iv) Make assumptions about any missing data.

1. (a) Give the reasons for the popularity of the Python language. (CO1)

OR

(b) What are the different types of data in Python ? Give examples of each.
(CO1)

2. (a) What are the different types of operators in Python ? Give examples.
(CO1)

OR

(b) Give an example of a code that inserts elements in a list and updates the same. Print the new list in reverse order. (CO1)

P. T. O.

(2)

OR

- (b) "I will learn and improve only if I am unhappy. If I become happy, my learning will stop." Explore the validity of this viewpoint. (CO2, CO3)
4. (a) "Right understanding + Relationship = Mutual fulfilment; Right understanding + Physical facilities = Mutual prosperity." Illustrate the above with *two* examples for each. (CO1, CO3)

OR

- (b) "Physical facilities are necessary and complete for animals, while they are necessary but not complete for humans." Comment. (CO1, CO3)
5. (a) Human being is the co-existence of the self and the body. What are the three specific distinguishing attributes of the self and the body ? Explain with examples. (CO2)

OR

- (b) Do you feel that you have some pre-conditionings ? How do you evaluate them ? (CO1, CO4)